

Individual Honours Project

WEEKLY REPORT

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I. SUMMARY

- This is the report of my project “BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration.”
- The target was to make EEG-based motor imagery calibration more effective and easier to apply outside the laboratory.
- Current progress:
 - Learned to make games using Game Maker Studio 2
 - Experienced in using EEG headband to collect research data
 - Investigated several EEG headbands on the market
 - Tested the Emotiv SDK to stream data from their headband
 - Having 2 game ideas for Motor Imagery calibration

II. What I have done

Device Investigation and Selection

- Investigated and tested 4 EEG devices: Brain-Life Focus+, Muse S Athena, Emotiv EPOC X, BrainBit Headband
- Selected **Emotiv EPOC X** as the main experimental device, as it:
 - Provides the built-in Motor Imagery features for my testing
 - Supports creating virtual device -> not depending on borrowing the headset

II. What I have done

Emotiv Software Development Kit

- I contacted Emotiv support to obtain more information about SDK capabilities and licensing. Two licensing options introduced:
- **Basic License** – access to processed cognitive metrics (0.1 Hz)
 - **Performance Metrics:** stress, engagement, interest, excitement, attention, relaxation
 - **Mental Commands:** trained mental actions like push, pull, lift, drop, left, right
 - **Facial Expressions:** blink, smile, wink, etc.
 - **Motion Data:** accelerometer and head movement
- **Premium License** – access to raw EEG signals and higher data streaming frequency (~2 Hz)
- After our discussion, they offered me access to a 6-month **basic license + raw EEG data permission** for research and study.

II. What I have done

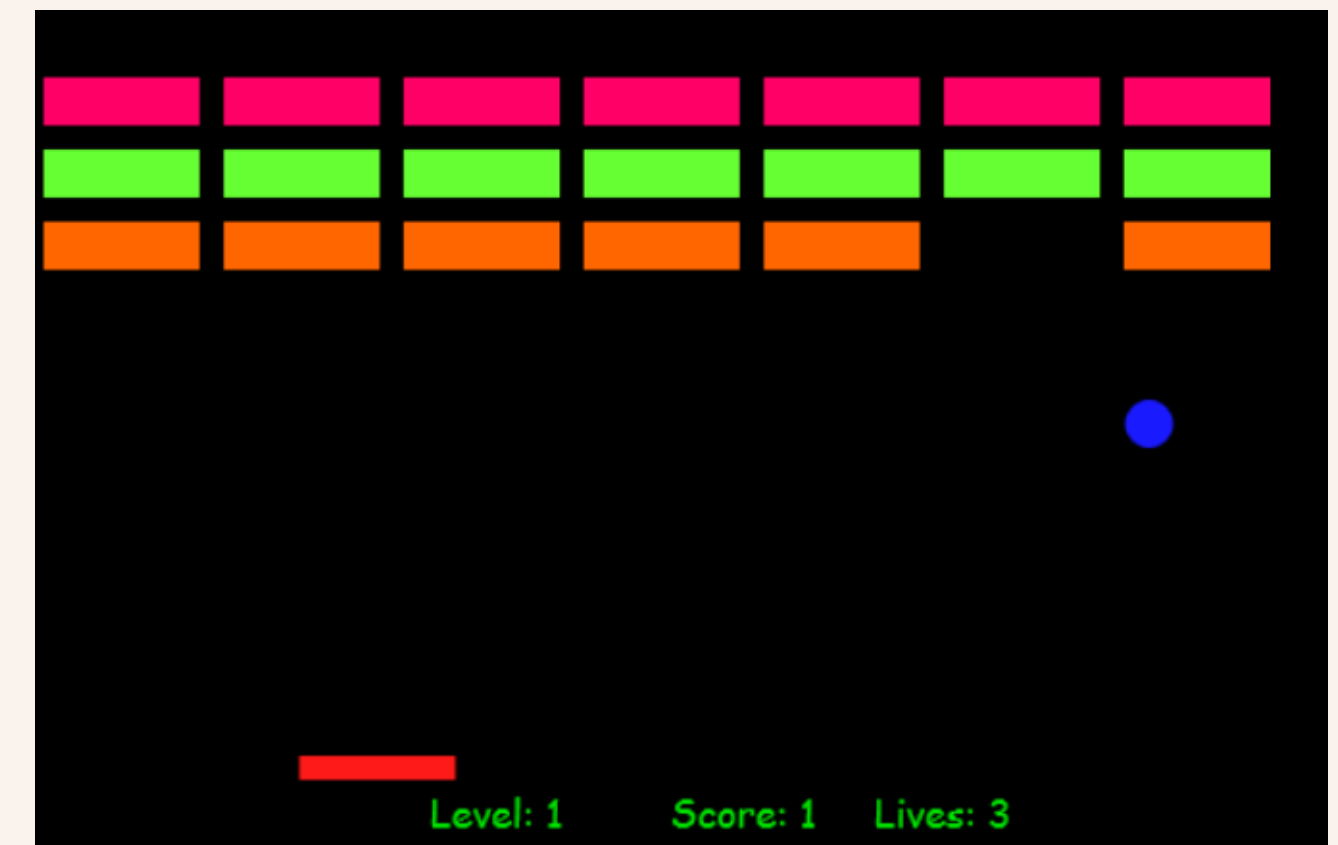
Emotiv Software Development Kit

- I have successfully accessed the **processed metrics** and **raw EEG data** in real time from the headband using a computer running Python scripts.
- I am currently continuing to discuss with Emotiv to get the 2Hz signal for better real-time experience.

II. What I have done

Game Idea

- Initially, I was considering creating a cool story-driven game with dynamic difficulty based on stress levels and controlled by Motor Imagery.
- But increased complexity could take up too much time for players, and our goal was to develop a more effective gamified calibration method.
- So I decided to keep the design simple to focus on improving MI calibration. My two simple game ideas:
 - Tree-cutting game
 - Brick break game



III. Current Difficulties

- Time flies like an arrow.
- As I gain more hands-on experience with different EEG devices, I realize that many technical challenges have already been addressed in existing research. This made my confidence fluctuate a bit!
- Should I use the public Motor Imagery dataset as a part of my training dataset? What should be the ratio between the public dataset and my collected data?

III. Next Steps

- Implement the program to process raw EEG data into features
- Implement the streaming script for those features to the game engine.



**THANK YOU
FOR READING!**

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